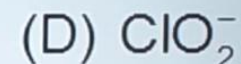
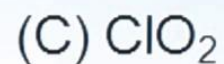
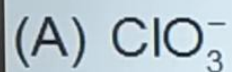
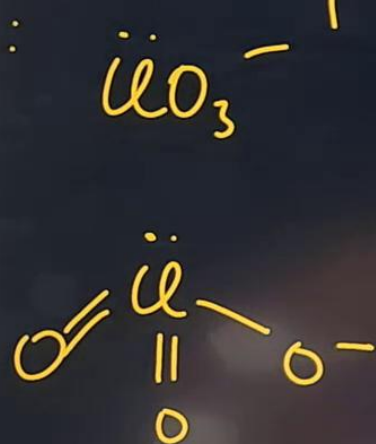


Species having minimum 'Cl-O' bond length is :



Solⁿ:



$$\text{B.O} = \frac{5}{3}$$

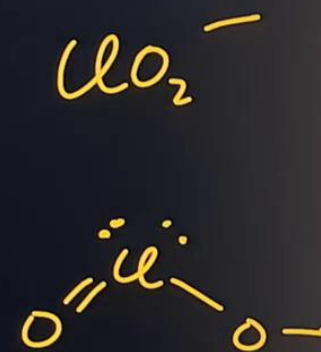
X



$$\text{B.O} = 2$$



$$\text{B.O} = 2$$

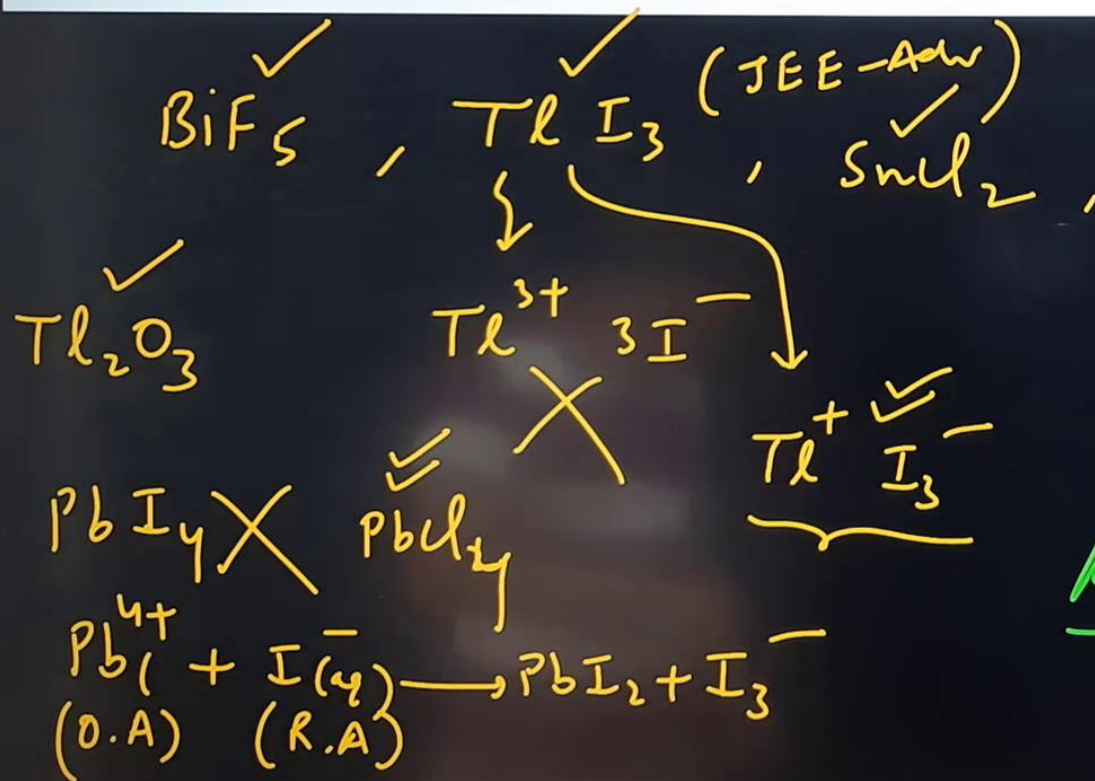


$$\text{B.O} = 1.5$$

X

Ans: (B)

How many of the following compounds are found to exist ?



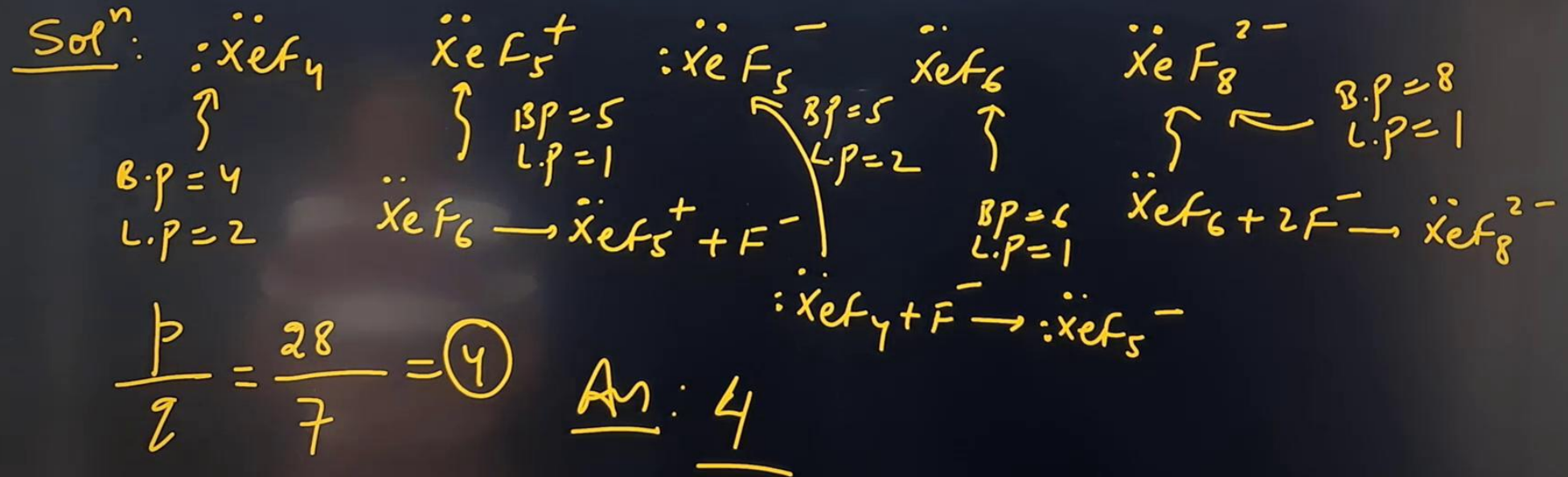
$\hookrightarrow$ Inert Pair effect.
 F, O form most strong bonds.

Ans: 6

Consider following compounds A to E :

- (A) XeF_n (B) $\text{XeF}_{(n+1)}^+$ (C) $\text{XeF}_{(n+1)}^-$
 (D) $\text{XeF}_{(n+2)}$ (E) XeF_{2n}^{2-}

If value of n is 4, then calculate value of " $p \div q$ " here, ' p ' is total number of bond pair and ' q ' is total number of lone pair on central atoms of compounds (A) to (E)



Consider the following species and find out total number of species which are polar and can act as Lewis acid

CCl_4 , CO_2 , SO_2 , AlCl_3 , HCHO , SO_3 , SiCl_4 , BCl_3 , CF_4

Solⁿ: CCl_4 X, CO_2 X, SO_2 ✓, AlCl_3 X, HCHO ✓, SO_3 X

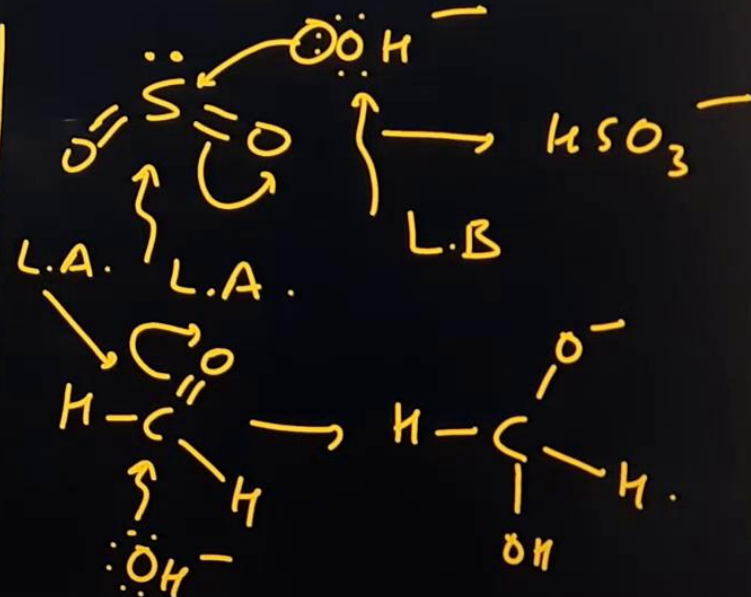
SiCl_4
X

BCl_3
X

CF_4
X

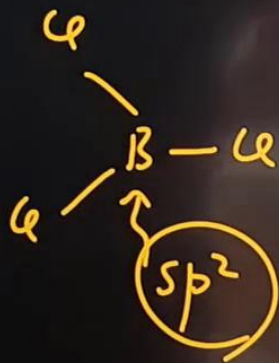
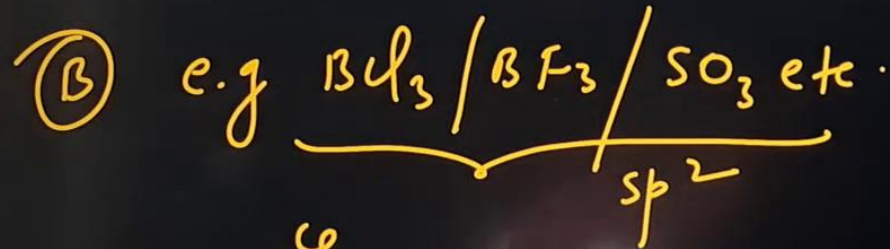
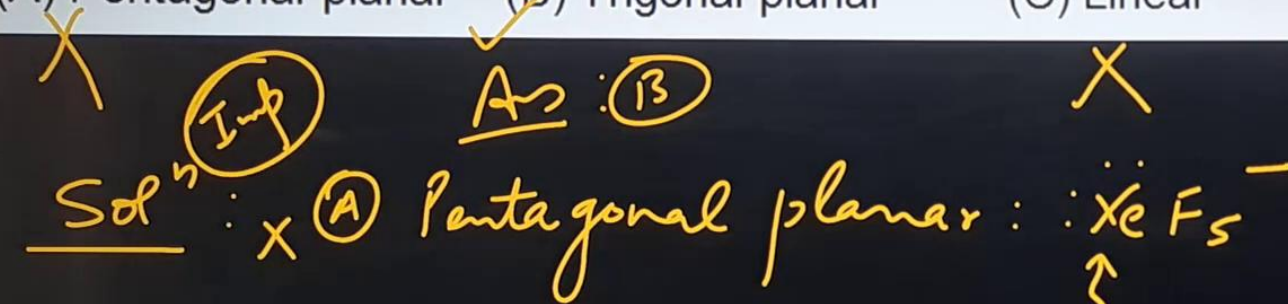
✓ : Polar

X : Non-polar

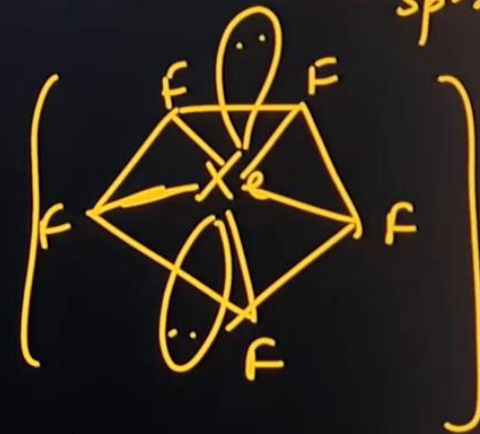
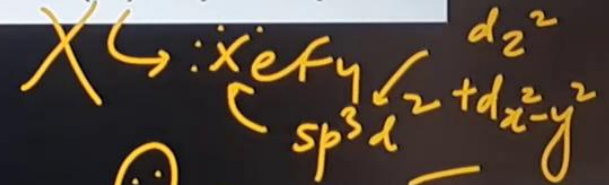
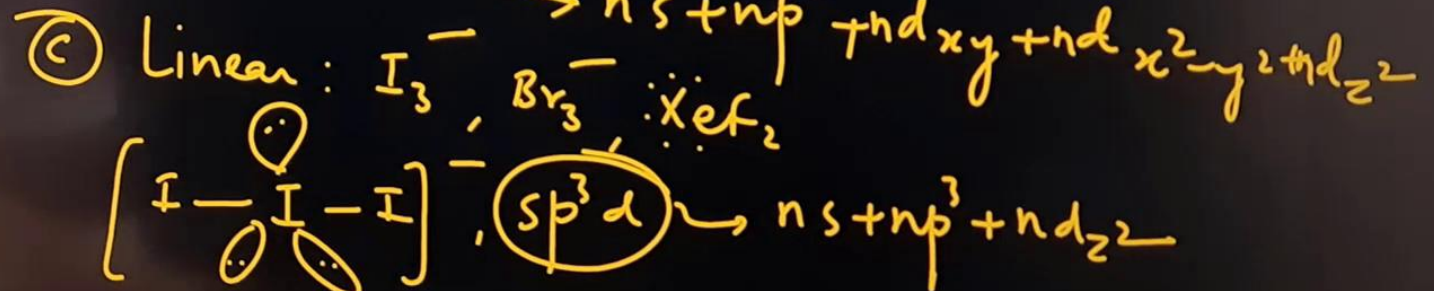


In which of the following molecular shape d_{z^2} -orbital must not be involved in bonding?

- (A) Pentagonal planar (B) Trigonal planar (C) Linear (D) Square planar



$s \cdot N = 5 + 2 = 7$
 $\text{sp}^3 d^3$



Which of the following orbital can not form π as well as δ -Bond?

(A) d_{xy}

(B) d_{z^2}

(C) $d_{x^2-y^2}$

(D) d_{yz}

Solⁿ: Ans: (B) : d_{z^2} forms σ -bond.

z-axis (internuclear axis)

$d_{x^2-y^2} + d_{x^2-y^2}$: δ -bond (delta bond as in $[\text{Re}_2\text{Cl}_8]^{2-}$)

$d_{xy} + d_{xy}$: δ -bond.

$d_{xz} + p_x$: π -bond.

$d_{yz} + p_y$: π -bond.

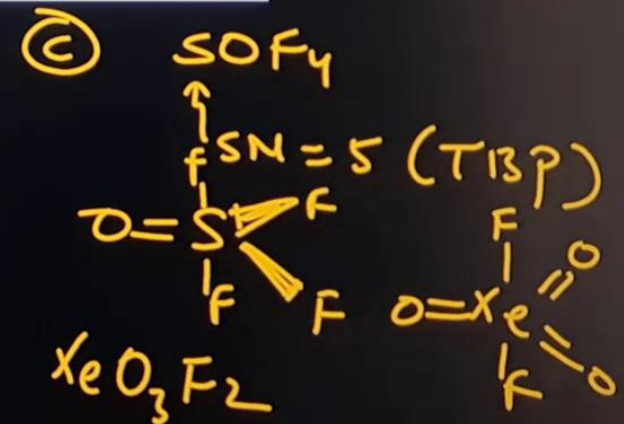
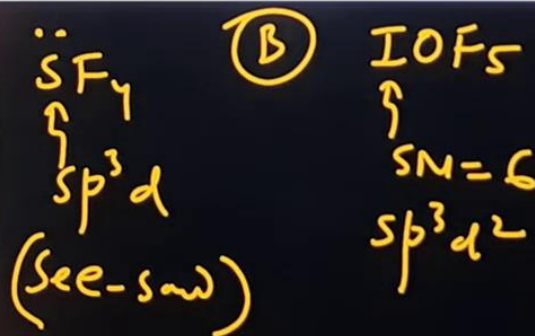
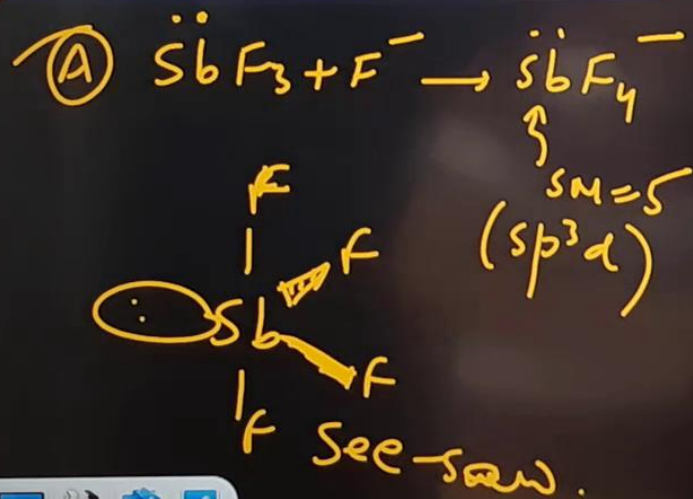
} d_{z^2} forms only sigma bond.

According to VSEPR model, molecules adopt geometries in which their valence electron pairs position themselves as far from each other as possible. The VSEPR model considers double and triple bonds to have slightly greater repulsive effects than single bonds because of the repulsive effect of π -electrons. However the lone pair creates the maximum repulsive effect.

Which of the following statement is false?

- (A) SbF_4^- and SF_4 are isostructural
- (B) In IOF_5 the hybridization of central atom is $sp^3 d^2$
- (C) Double bond(s) in SOF_4 and XeO_3F_2 , is/are occupying equatorial position(s) of their respective geometry
- (D) None of these

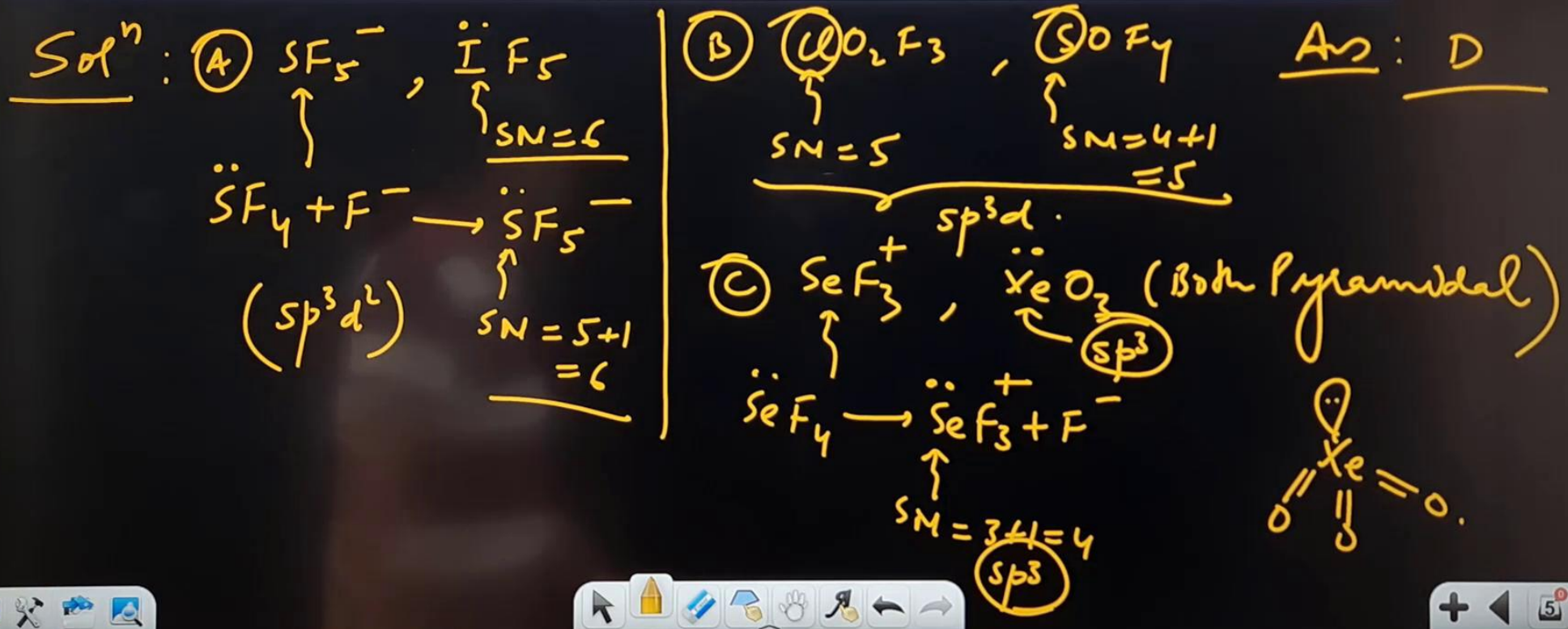
Ans: (D)



According to VSEPR model, molecules adopt geometries in which their valence electron pairs position themselves as far from each other as possible. The VSEPR model considers double and triple bonds to have slightly greater repulsive effects than single bonds because of the repulsive effect of π -electrons. However the lone pair creates the maximum repulsive effect.

Which of the following does not represent the isostructural pair?

- (A) SF_5^- and IF_5 (B) ClO_2F_3 and SOF_4 (C) SeF_3^+ and XeO_3 (D) None

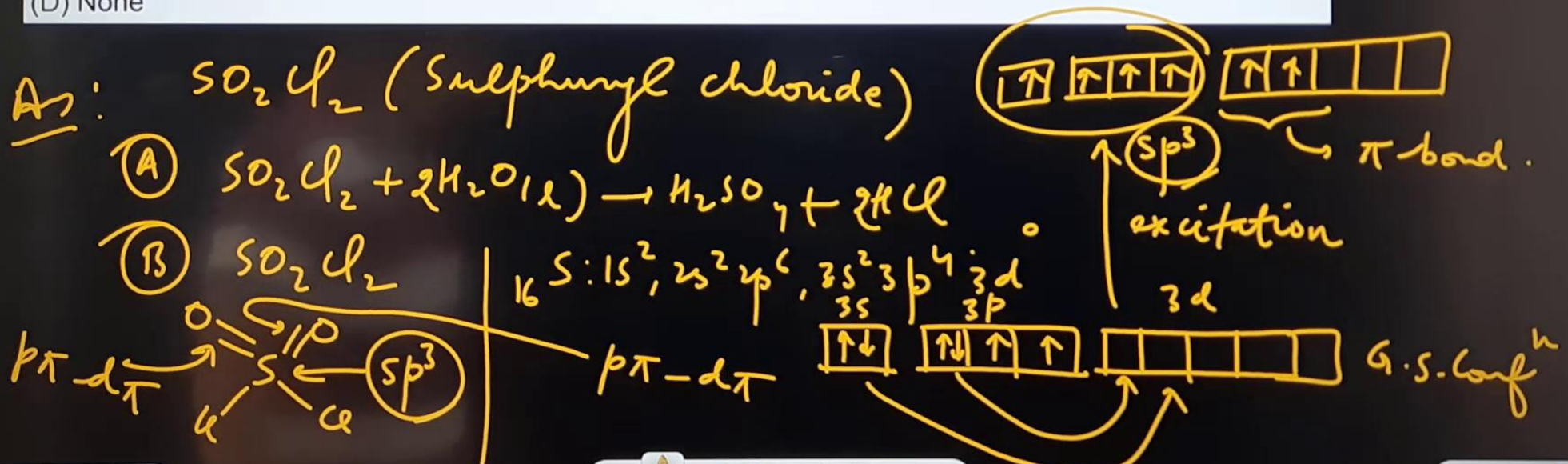


According to VSEPR model, molecules adopt geometries in which their valence electron pairs position themselves as far from each other as possible. The VSEPR model considers double and triple bonds to have slightly greater repulsive effects than single bonds because of the repulsive effect of π -electrons. However the lone pair creates the maximum repulsive effect.

Select the incorrect statement with respect to SO_2Cl_2 molecule :

- (A) It gives H_2SO_4 and HCl on hydrolysis at room temperature
- (B) It has two $d\pi-p\pi$ bonds between S and O bonded atoms
- (C) It is a polar molecule
- (D) None

Ans: (D) ←



Ionic compounds in general do not possess :

(A) high melting points and non-directional bonds

(B) high melting points and low-boiling points

✓ (C) directional bonds and low-boiling points

(D) high solubilities in polar and non-polar solvents

Ans: C

Covalent bonds: directional.
Ionic bond
or electrovalent bond:
Non-directional.

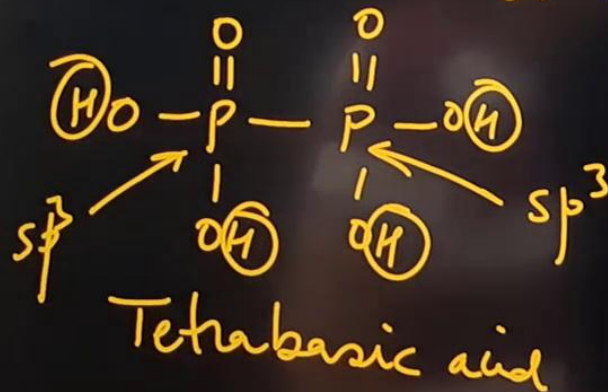
Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with some entries of Column-II. One or more than one entries of Column-I may have the matching with the same entries of Column-II.

Column-I	Column-II
(A) Hypo phosphoric acid	(p) All hydrogen are ionizable in water
(B) Pyro phosphorous acid	(q) Lewis acid
(C) Boric acid	(s) sp^3 hybridised central atom
(D) Hypo phosphorous acid	

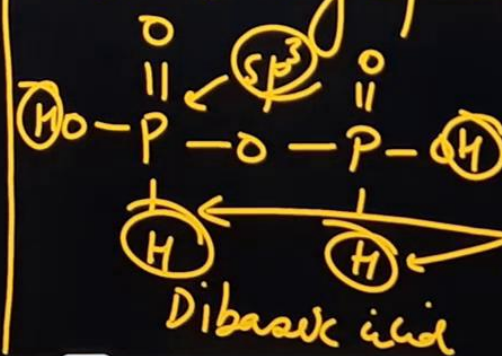
$A \rightarrow P, S$
 $B \rightarrow S$
 $C \rightarrow q, r$
 $D \rightarrow r, s$

sp^2
 H_3BO_3
 Lewis,
 monobasic
 acid

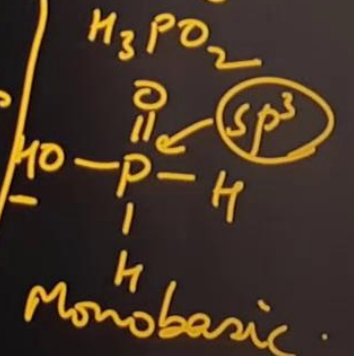
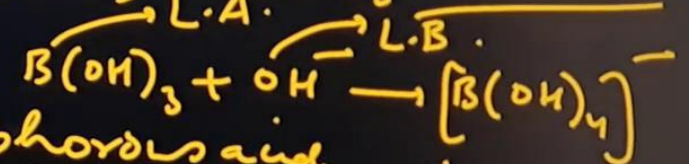
$H_4P_2O_6$: Hypophosphoric acid



$H_4P_2O_5$: Pyrophosphorous acid



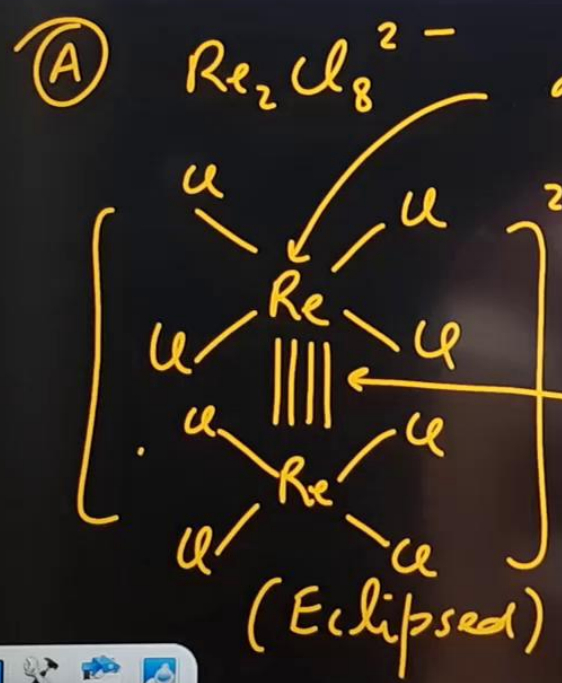
H_3BO_3 or $B(OH)_3$: Boric



Column-I			Column-II	
(A)	$\text{Re}_2\text{Cl}_8^{2-}$	(p)	$\pi-\pi$ bonding	
(B)	NO_3^-	(q)	$\pi-d\pi$ bonding	
(C)	SO_4^{2-}	(r)	$d\pi-d\pi$ bonding	
(D)	SO_3	(s)	δ -bonding	

V. Good

Delta bond is found in,
 $\{\text{ReCl}_8^{2-}, \text{Os}_2\text{Cl}_8^{2-}, \text{Mo}_2\text{Cl}_8^{4-}\}$



$dsp^2 ((n-1)d_{x^2-y^2} + ns + np_x + np_y)$

$\text{O}=\text{S}=\text{O}$ (sp³)

B.O between two Re atoms = 4 (quadruple bond)

1 σ (sigma)
 2 π (pi)
 1 δ (delta)

$\text{O}=\text{S}=\text{O}$ (sp³)

$\pi-\pi$
 $\pi-d\pi$
 d_{z^2}
 d_{xz}, d_{yz} (delta bond)
 dxy orbital forms